Supplementary Material for: Minimal Operational Distinguishability Rank Four and Structural Non-Factorisation of Composite Systems

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About this Supplementary Material

This document accompanies the paper "Minimal Operational Distinguishability Rank Four and Structural Non-Factorisation of Composite Systems" by A. M. Tishin. Sections are numbered S1–S14 and Appendices A–C; section, theorem and equation numbers without the prefix S refer to the main paper. Reference numbers [n] refer to the reference list at the end of this document. Repository: https://github.com/alexandermtishin-art/Minimal-Operational-Distinguishability-Rank-Four.

S1 Operational memory and metastability

A stored correlation is represented by a distinction detectable through an admissible procedure. The operational description does not identify its microscopic storage mechanism. Persistence is relative to the declared set of continuations: it requires that none of the allowed erasure procedures removes the distinction.

The magnetic phases of Fe–Rh provide an illustration of distinguishable macroscopic states [1]. Magnetisation and caloric measurements can distinguish the phases in suitable experimental regimes. A quantitative account of their preparation dependence requires the sample, control protocol, temperature range and available operations to be specified.

No universal erasure temperature or exclusion of heating from the procedure monoid is assumed here. Heating is excluded only if the chosen operational control domain explicitly excludes it. Thermodynamic barriers and kinetics may then explain why a distinction is stable within that domain; the abstract monoid does not replace that explanation.

Local operations cannot create entanglement from a separable state, but they can reduce or destroy existing entanglement [2]. Thus locality alone does not imply persistence. Bell nonlocality is also distinct from entanglement [3]. The present domain and procedure criteria describe structural non-factorisation; a claim about quantum entanglement additionally requires a specified quantum realisation.

S2 From distinguishability to information: the logical chain

S2.1 Motivation

The construction of §2.1–2.6 establishes the existence of multiple distinguishable states and their local organizational structure. This section makes explicit the logical chain connecting distinguishability to information. The chain is organised in six steps, but their logical status is not identical: some are primitives, some are consequences of the stated structural assumptions, and some are operational definitions. The purpose of the section is to keep these statuses explicit rather than to claim that every step is a theorem of the preceding one.

S2.2 Step 1: distinguishable states exist

There exist at least two physically distinguishable states $s, s' \in S$ with $s \nsim s'$. This is the foundational operational primitive. The present work asks: what structural conditions are required for this to be the case?

S2.3 Step 2: multiple distinguishable states exist

The main result of §3 constrains the minimum independent procedure rank of the stated structural class: C4b gives $n \geq 4$, an explicit witness achieves $n = 4$, and C3 selects that minimum for the fundamental carrier. This does not create multiplicity from consistency alone; the existence of operationally distinguishable states is already part of the primitive setting fixed in Step 1.

S2.4 Step 3: local accessibility

Given $n = 4$ (Theorem 3.1; under C4b, $\mathrm{rank}\, A([s]) = 4$), there are four independent composition-irreducible procedure directions in the C3-selected fundamental local carrier. This does not imply four distinct endpoint classes are directly reachable: several independent procedures may share an endpoint while differing on the full forward domain. Non-trivial local accessibility itself is supplied by C1 and the transition structure of §2.4.2.

S2.5 Step 4: interaction

A typed morphism in C_{D} represents an admissible operational transformation between covering classes. Calling such a transformation an ‘interaction’ is only operational shorthand for the existence of a procedure with a registered effect; no dynamical force, causal influence, energetic exchange, or temporal evolution is derived from the morphism structure alone.

S2.6 Step 5: measurement

A measurement is represented by an admissible typed morphism whose resulting complete operational state includes a registered outcome or record that distinguishes the relevant alternatives. The framework therefore supplies a structural representation of measurement, while the statistical law assigning probabilities to different outcomes remains additional structure.

S2.7 Step 6: information

A measurement record carries operational information insofar as it discriminates among finitely many distinguishable outcome records allowed by Axiom F. No probability distribution or Shannon entropy is required merely to state which distinguishable record occurred; quantitative information measures require further structure.

Scope. The core supplies a language for procedures, records and composite domains. It does not identify structural non-factorisation with quantum entanglement. Main-text §3.4.4 adds a formal state-set linearisation, while §§3.4.5–3.4.6 give a quantum-operation example and a deterministic record criterion. Their additional data are stated explicitly; no outcome probability law is assumed in the core.

S2.8 Summary of the chain

Step 1: distinguishability exists as an operational primitive. Step 2: the stated structural class constrains the minimum independent operational rank. Step 3: C1 supplies non-trivial local accessibility, while rank four counts independent irreducible procedure directions rather than distinct target classes. Step 4: typed morphisms represent operational transformations. Step 5: measurement is represented by an outcome-registering typed morphism. Step 6: a distinguishable outcome record carries operational information. The primitive assumptions, derived rank statement, and operational definitions are thereby kept logically distinct.

S2.9 Relationship to Shannon information theory

The definition of §S2.7 is more primitive than Shannon$'s$ [4]: it requires only distinguishable outcomes, not a probability distribution. Once probabilities on outcome classes are supplied, Shannon entropy provides a quantitative information measure on the resulting distributions. The present operational notion is logically prior to that probabilistic layer, but it is not claimed to subsume all of Shannon information theory. [4]

S3 Auxiliary consistency analysis: the complementary-support Spin construction

Proposition S3.1 (post-selection $\mathrm{Spin}(4)$ algebraic structure). Once $n = 4$ has been established independently by the counting theorem (Theorem 3.0, Corollary 3.0), and a positive quadratic form has been introduced on V_{G}, the associated Clifford/spin structure admits the chiral factorisation $\mathrm{Spin}(4,\mathbb{C}) \cong \mathrm{SL}(2,\mathbb{C})_{\mathrm{L}} \times \mathrm{SL}(2,\mathbb{C})_{\mathrm{R}}$. This is a property of the spinorial structure built from V_{G} via its Clifford algebra, not a splitting of V_{G} itself, and not a realisation of C4a or C4b: the automorphism group $\rho_{\mathrm{G}}(\mathrm{Aut}(P([s]))_{\mathrm{G}}) \subseteq O(n)$ is not asserted to realise this factorisation on V_{G} directly, nor is any such splitting of V_{G} claimed.

S3.1 Method of analysis

Theorem 3.0 and the P_4 witness have already settled the admissibility bound relevant to Theorem 3.1. The present section does not test C4a or C4b and does not contribute to the proof of $n_{\min} = 4$. It asks only at which dimensions the additional complementary-support representation-theoretic construction is algebraically available — an auxiliary comparison, not an admissibility test.

We flag the logical status of this auxiliary analysis explicitly. The exclusion n < 4 in the main theorem has already been proved by the representation-free counting bound, so no Spin-group classification is needed for Theorem 3.1. The case discussion below asks a different question: for which n does the connected complex spin group itself admit a non-trivial direct-product decomposition into two positive-dimensional connected normal factors? For that auxiliary question it is appropriate to use the standard low-dimensional identifications of $\mathrm{Spin}(n,\mathbb{C})$. These group-theoretic facts are recorded as post-selection structure only and are not transferred to an arbitrary finite local stabiliser subgroup.

No physical spacetime geometry is assumed in this auxiliary analysis. The primary operational objects remain the indistinguishability classes $[s], A([s])$, its rank, admissible procedures, and

the consistency conditions of §3.1–3.2; only after the counting result is fixed do we add an auxiliary positive quadratic form and the associated Clifford/spin carrier for the separate post-selection comparison below.

S3.2 Case $n = 1$

At $n = 1$, the counting theorem excludes C4b. Separately, the complex spin group has no positive-dimensional connected factors, so the auxiliary two-factor construction is unavailable.

Verdict: the complementary-support form is unavailable at n = 1.

S3.3 Case $n = 2$

Admissibility at $n = 2$ is already excluded by Theorem 3.0. For the separate auxiliary product-decomposition question, $\mathrm{Spin}(2,\mathbb{C})$ is a one-complex-dimensional torus (isomorphic to $\mathbb{C}\times$). A connected one-dimensional complex Lie group cannot be expressed as a direct product of two positive-dimensional connected normal factors, because the dimensions of such factors would have to add to one. Hence the auxiliary product structure is unavailable at $n = 2$. This statement concerns the connected complex spin group itself; it makes no claim that every finite subgroup of $O(2)$ is abelian and requires no orientation-preserving assumption on the operational local stabiliser.

The auxiliary two-factor construction is unavailable at dimension two. This does not assert that every finite subgroup of the orthogonal group is Abelian.

S3.4 Case $n = 3$

Admissibility at $n = 3$ is likewise already excluded by the counting theorem. For the auxiliary group question, $\mathrm{Spin}(3,\mathbb{C}) \cong \mathrm{SL}(2,\mathbb{C})$. Its Lie algebra $\mathfrak{sl}(2,\mathbb{C})$ is simple, so the connected group is almost simple modulo its finite centre and has no decomposition into two positive-dimensional connected normal factors. Therefore the auxiliary product structure is unavailable at $n = 3$. Again, this is a statement about the connected ambient Spin group, not an assertion that a finite local stabiliser must inherit the normal-subgroup structure or irreducibility of $\mathrm{SL}(2,\mathbb{C})$.

Finite-subgroup objection. A finite subgroup of an almost-simple ambient Lie group can certainly have proper normal subgroups or reducible representations. The auxiliary property being classified, however, is explicitly a product decomposition of the connected complex Spin

group itself, not of every finite subgroup that may sit inside it. Thus no inference from ambient simplicity to finite-subgroup simplicity is used. The local-stabiliser representation and the Spin-group factorisation are separate constructions on different carriers, as emphasised again in §S4 and Appendix A.

The operationally relevant finite group $\rho_{\mathrm{G}}(\mathrm{Aut}(P([s]))G)$ remains governed only by the local-stabiliser theorems: it acts faithfully on the chosen frame and can be conjugated into $O(n)$ by finite averaging. Nothing in the $n = 3$ auxiliary case assumes that this finite group equals $\mathrm{Spin}(3,\mathbb{C})$, is dense in it, or shares its normal-subgroup structure. The ambient classification is used only to answer the explicitly auxiliary question just stated.

Accordingly, the auxiliary connected-product property is unavailable at $n = 3$ for the standard and representation-independent reason that $\mathrm{Spin}(3,\mathbb{C})$ is almost simple. The main inadmissibility of $n = 3$ remains the independent counting result of Theorem 3.0; no new physical postulate is introduced by this auxiliary observation.

Scope of the complementary-support language. A description of the auxiliary product through 'observer' and 'state' support sectors on a representation space would be stronger than the group-theoretic fact actually needed here and is not derived from C4a or C4b. In the present architecture the formal auxiliary statement is only the existence, at $n = 4$, of two non-trivial commuting connected normal factors in the complex spin group. Any identification of those two factors with physical observer and state sectors requires additional structure and is not used in the proof of $n_{\min} = 4$.

The low-dimensional minimum of this auxiliary group property is therefore four. At $n = 1$ there is no non-trivial positive-dimensional Spin group to split; at $n = 2$ the connected group has complex dimension one; at $n = 3$ it is almost simple; and at $n = 4$ one has the explicit factorisation $\mathrm{Spin}(4,\mathbb{C}) \cong \mathrm{SL}(2,\mathbb{C})_{\mathrm{L}} \times \mathrm{SL}(2,\mathbb{C})_{\mathrm{R}}$. This establishes the first occurrence of the auxiliary product decomposition without invoking a decomposition of the operational generator space.

This conclusion should not be read backwards. It does not say that every subgroup at $n = 4$ factorises, nor that no finite subgroup at $n = 2$ or $n = 3$ has an internal direct-product structure. It says only that the connected complex Spin group used as the auxiliary classifying carrier first possesses the stated positive-dimensional normal-factor decomposition at $n = 4$.

The distinction is useful because it prevents the auxiliary discussion from contaminating the main theorem. The representation-free counting route excludes $n = 1,2,3$ for C4b monoids. The Spin classification merely notes that a familiar chiral product becomes available once the independently selected value $n = 4$ is inserted into the auxiliary Clifford/spin construction.

Verdict for the auxiliary construction: the connected complex Spin product is unavailable for $n = 1,2,3$ and first available at $n = 4$. This is not an additional admissibility criterion and is not used to obtain Theorem 3.1.

S3.5 Case $n = 4$

Admissibility at $n = 4$ has already been established by the P_4 witness (Theorem 3.0, Corollary 3.0, the counting route). We ask only whether the auxiliary Spin construction is available at that already-selected value.

The auxiliary product-decomposition construction is algebraically available at $n = 4$. The counting route (Theorem 3.0, Corollary 3.0) has already established the four-generator minimum independently, so nothing in this subsection is used to prove or confirm that value. The additional question is purely representation-theoretic: after a positive quadratic form and its Clifford/spin carrier are introduced, does the associated complex spin group admit two non-trivial commuting connected normal factors? At $n = 4$ it does, through $\mathrm{Spin}(4,\mathbb{C}) \cong \mathrm{SL}(2,\mathbb{C})_{\mathrm{L}} \times \mathrm{SL}(2,\mathbb{C})_{\mathrm{R}}$. This factorisation lives on the auxiliary spinorial carrier and is not a decomposition of the generator space VG, not a theorem that the full categorical $\mathrm{Aut}(C_{\mathrm{D}})$ factorises, and not an operational derivation of an observer/state split. The labels 'observer' and 'state' are therefore interpretive labels for the two auxiliary factors unless an additional physical bridge is supplied. Appendix A.4 records the precise scope: the construction is a post-selection algebraic structure available once $n = 4$ is already fixed, while the local-stabiliser representation of Theorems A.2–A.3 remains a separate finite-group statement.

Verdict: the complementary-support form is first available at $n = 4$. This is a fact about the auxiliary $\mathrm{Spin}(n,\mathbb{C})$ construction on its own carrier (the half-spin modules S^+, S^-), recorded for its own interest; it neither confirms nor establishes the value $n = 4$, which is secured unconditionally, and independently of this construction, by the counting route.

S3.6 Cases $n = 5, 6, 10, 11$ and higher

Higher values $n \geq 5$ are not excluded by the counting lower bound of Theorem 3.0, but they are non-minimal whenever they satisfy the stated structural conditions; their existence is not required by Theorem 3.1.

However, C3 — minimality, restated in §3.2.6 as the requirement that n take the smallest value achieved within the C1, C2, C4a, C4b class — is violated by definition: once $n = 4$ is established as the smallest such n by the main-text counting theorem and explicit witness (Theorem 3.1), any $n \geq 5$ is non-minimal as a direct consequence of what minimality means, not as a separate claim requiring proof for each individual $n \geq 5$. By the Atomic-Generation Theorem under C4b, the same conclusion transfers to $d_{\text{eff}} \geq 5$.

Higher generator ranks are allowed by C4b. They are not identified here with extra spatial dimensions, compactification scales or hidden geometric coordinates.

What the present result establishes is the following: $n = 4$ is confirmed to satisfy all four requirements and is the unique minimal admissible value selected by C3; $n = 1, 2, 3$ are excluded by the counting bound of Theorem 3.0, while the auxiliary complementary-support construction is separately unavailable at these values (§S3.4). Higher-rank structures, if physically realised, are non-minimal under C3. If, in addition, such a structure admits an admissible reduction to a target satisfying C1, C2, C4a, and C4b and selected there by C3, the target local monoid has $d(P) = 4$, and under C4b its local operational rank is also four (Corollary 3.2). The existence of such a reduction is not established here (Open Problem B.0, Appendix B). Verdict: $n \geq 5$ is non-minimal under C3, not inconsistent with C1, C2, C4a, and C4b. Whether any higher-rank structure admits the required operational reduction remains open.

S4 Minimal sufficient data for reconstructing the $\mathbf{Spin}(4)$ post-selection construction

This section supplies the auxiliary frame representation and, after a separate quadratic-form choice, the Clifford/spin construction. None of these constructions is used in the counting or atomic-generation proofs.

Step 1 (finite generation). Let $G = \{g_1, ..., g_n\}$ be a generating set of $P([s])$ achieving the minimum cardinality $n := d(P([s]))$ (§2.4.7a). Axiom F (finite resolution of outcomes) alone does not force G to be finite — a system could in principle admit infinitely many independent procedures, each with finitely many outcomes. Finiteness of n requires the additional Axiom G

(local finite generation): the monoid $P([s])$ is finitely generated. Given Axiom F and Axiom G together, n is finite (Theorem A.1). The representation space $\mathbb{V}([s])$ is defined directly as $\mathbb{R}$G, the free vector space on this n-element generating set, so that $\dim \mathbb{V}([s]) = n$ holds by construction, not by a quotient of the full (generically infinite) monoid $P([s])$. $n = d(P([s]))$ is, by definition of the minimum, a single well-defined invariant of $[s]$ — no further assumption is needed for this, and it is not the content of the rank–generator equality, which addresses a separate question: whether this representation invariant n coincides with the independently defined operational invariant $\mathrm{rank}\, A([s])$ — whether two different combinatorial notions of size agree for these monoids, not whether n itself is well-defined (§2.4.7a).

Step 2 (the local frame stabiliser). For a minimum generating set G, let $H_G = \{\phi \in \mathrm{Aut}(P) : \phi(G) = G\}$. Its permutation action extends linearly to $V_G = \mathbb{R}^G$. It is faithful: fixing all generators fixes every word in them. Hence H_G embeds in the finite group $\mathrm{Sym}(G)$. In the basis indexed by G, the standard Euclidean inner product is already invariant; finite averaging gives the same existence statement starting with any positive-definite inner product. This is a local monoid representation, not a representation of arbitrary categorical automorphisms.

A potential concern is whether this lands in $O(V_\mathrm{G})$ only because of the particular starting form chosen, with a different choice perhaps giving a different kind of group. It does not: by Sylvester's law of inertia, every positive-definite bilinear form on an n-dimensional real vector space is congruent to every other, so the isometry group obtained is, as an abstract group, isomorphic to $O(n)$ regardless of which positive-definite form one starts the averaging from. The averaging step is forced to land in $O(V_\mathrm{G})$, not merely a natural or convenient place for it to land (Appendix A, Step 3).

A related concern is whether this construction secretly depends on the choice of minimal generating set G, making the geometry an artefact of linearisation rather than a property of $P([s])$ itself. We do not claim this dependence is absent in general: different minimum frames G and G′ define isomorphic vector spaces of the same dimension, but no canonical identification of their stabiliser representations ρ_G and $\rho_\mathrm{G}{}'$ is assumed here, and their stabilisers $\mathrm{Aut}(P([s]))_\mathrm{G}$ and $\mathrm{Aut}(P([s]))_\mathrm{G}{}'$ need not coincide or be conjugate as subgroups of $\mathrm{Aut}(P([s]))$. What is basis-independent is the dimension n itself (Theorem A.1) and, for a fixed choice of G, the conjugacy class of ρ_G within $\mathrm{GL}(V_\mathrm{G})$ under change of orthonormal basis of V_G — not any identification across different choices of G.

A potential concern is whether $\mathrm{Aut}(P([s]))$, being built from a purely algebraic and possibly infinite structure, could act as a dense, non-closed subgroup — the kind for which compactness arguments and Haar measure would not apply. This concern does not arise for the stabiliser construction: $\rho_{\mathrm{G}}(\mathrm{Aut}(P([s]))_{\mathrm{G}})$ is shown above to be finite by direct construction (it factors through the permutation group of the n generators, which has only $n!$ elements), not merely assumed or argued to be compact after the fact. There is accordingly no dense or non-closed case remains: finiteness is established directly, before any topological notion is needed.

Note on a related but different objection. A natural-sounding but ultimately misdirected criticism is that establishing the structure of $\rho_{\mathrm{G}}(\mathrm{Aut}(P([s]))_{\mathrm{G}})$ requires invoking Cartan's closed-subgroup theorem on an a priori topologised ambient group — which would indeed reintroduce the topology this construction is built to avoid. The construction above sidesteps this entirely: finiteness of $\rho_{\mathrm{G}}(\mathrm{Aut}(P([s]))_{\mathrm{G}})$ is established directly, by exhibiting it as a subgroup of the (finite) permutation group of the n generators, before any topology is introduced. No ambient Lie group, closed-subgroup theorem, or Cartan-type argument is invoked at any point.

Step 2b (orientation). For a specified generator-preserving observer subgroup, the determinant is multiplicative. Its image lies in $\mathrm{SO}(n)$ if every generating frame change has determinant plus one. This condition is not implied by finite-chain reachability and is not imposed in C4b. In particular, the atom transpositions of the Boolean witness are orientation reversing on their frame representation.

Algebraic reachability does not impose orientation preservation. A transposition of two frame generators has determinant minus one and can belong to the admissible observer group. Restriction to determinant plus one is an extra condition, and is not imposed in the counting theorem.

Step 2a (invariance of $n = d(P)$, used implicitly throughout). For any admissible observer transformation φ (§2.4.9 of the main text), n is the same at [s] and at φ([s]): conjugation by φ gives a monoid isomorphism $P([s]) \cong P(\varphi([s]))$, and monoid isomorphisms preserve minimum generating-set cardinality — $d(P([s])) = d(P(\varphi([s])))$ — since φ carries any generating set of $P([s])$ to a generating set of $P(\varphi([s]))$ of the same cardinality, and φ^{-1} carries any generating set of $P(\varphi([s]))$ back to one of $P([s])$ of the same cardinality, so the minimum over all generating sets on each side agrees. This is what makes $'n$ does not depend on the observer' precise, and underlies the construction of ρ_{G} in Step 2 as well as the formulation of C4

itself in §3.2; the operational statement that $\mathrm{rank}\, A([s])$ does not depend on the observer additionally invokes the rank–generator equality (Atomic-Generation Theorem, §2.4.5a).

Step 3 (faithfulness). An automorphism fixing a generating set pointwise fixes every monoid element, so the local frame representation is injective. This uses generation only; neither C3 nor a claim about its action on other local monoids is needed.

A potential concern is whether trivial action on one local monoid $P([s])$ really forces trivial action on the whole category $\mathrm{C_D}$ — could φ act trivially at $[s]$ yet non-trivially somewhere else entirely? This concern does not apply to the present step: injectivity here is a purely local statement, about the kernel of ρ_G restricted to its action on the single generating set G of $P([s])$ at the indistinguishability class $[s]$ under consideration — establishing $ker(\rho_G) = \{e\}$ for this representation. It makes no claim, and needs none, about how φ acts at other, unrelated classes $[s']$; no global or non-local argument is required, and none is invoked.

Step 4 (the case analysis at the level of the complexified spin group, and what it does and does not establish). To avoid introducing any real signature not derived in this paper, the auxiliary construction is realised at the level of the complexified group: a non-trivial decomposition of $\mathrm{Spin}(n, \mathbb{C})$ as a direct product of two commuting non-trivial normal subgroups is the post-selection structure recorded by Proposition S3.1. A clarification about the use of a continuous Lie group to classify a finite discrete group is warranted here. The group $\rho_{\mathrm{G}}(\mathrm{Aut}(P([s]))_{\mathrm{G}})$ established in Steps 1–3 is finite and discrete — this is the local stabiliser representation of §S4 Step 2, not a claim about the whole of $\mathrm{Aut}(\mathrm{C_D})$; it sits inside $O(n, \mathbb{R})$ as a finite subgroup.

A subgroup need not inherit the normal-subgroup structure of its ambient group. For example, finite cyclic subgroups can have nontrivial direct-product decompositions inside a connected group with simple Lie algebra. Thus the complex Spin classification below concerns the added ambient group only. All exclusions of generator ranks below four follow from the counting theorem.

The case-by-case classification of the connected complex spin groups is recorded here only for the auxiliary post-selection question. In the low-dimensional cases relevant to the comparison, $\mathrm{Spin}(2, \mathbb{C}) \cong \mathbb{C}\times$ is a one-dimensional complex torus; $\mathrm{Spin}(3, \mathbb{C}) \cong \mathrm{SL}(2, \mathbb{C})$ has simple Lie algebra $\mathfrak{sl}(2, \mathbb{C})$; $\mathrm{Spin}(4, \mathbb{C}) \cong \mathrm{SL}(2, \mathbb{C}) \times \mathrm{SL}(2, \mathbb{C})$ is the first case with two positive-dimensional connected normal factors; $\mathrm{Spin}(5, \mathbb{C}) \cong \mathrm{Sp}(4, \mathbb{C})$ and $\mathrm{Spin}(6, \mathbb{C}) \cong \mathrm{SL}(4, \mathbb{C})$ have simple Lie algebras, and $\mathfrak{so}(n, \mathbb{C})$ is simple for $n \geq 5$. $\mathrm{Spin}(8, \mathbb{C})$ has triality as an outer

automorphism phenomenon, but triality is not a direct-product decomposition of the group. Thus, apart from the exceptional split at $n = 4$, the connected complex groups in this range are almost simple modulo finite centre and admit no decomposition into two positive-dimensional connected normal factors. None of these ambient-group facts for $n \geq 5$ is needed to exclude those values from the C3-selected minimum, since the counting theorem already fixes the minimum independently.

Step 4a (uniqueness of the $n = 4$ decomposition, stated at the correct level). At the Lie-algebra level, the structure theorem for finite-dimensional semisimple Lie algebras gives a unique decomposition, up to exchange of factors, $\mathfrak{so}(4,\mathbb{C}) \cong \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$. For the simply connected complex spin group this integrates to the exact direct product $\mathrm{Spin}(4,\mathbb{C}) \cong \mathrm{SL}(2,\mathbb{C}) \times \mathrm{SL}(2,\mathbb{C})$, again unique up to exchange of the two simple factors. For finite central quotients in the same isogeny class, the Lie-algebra decomposition remains the same, but the corresponding connected normal subgroups can intersect in a finite central subgroup; therefore one has an almost-direct product or central isogeny rather than necessarily a literal direct product. The present result uses the exact product only for $\mathrm{Spin}(4,\mathbb{C})$ itself and does not select $\mathrm{Spin}(4,\mathbb{C}), \mathrm{SO}(4,\mathbb{C})$, a compact real form, or a Lorentzian real form as the physically relevant symmetry group.

Step 5 (Clifford convention and half-spin modules). Choose a positive-definite inner product on the four-dimensional frame space and the Clifford convention $v^2 = -\| v \|^2$. Then $\mathrm{Cl}_{0,4} \cong M_2(\mathbb{H})$ and $\mathrm{Cl}^0_{0,4} \cong \mathrm{Cl}_{0,3} \cong \mathbb{H} \oplus \mathbb{H}$. After complexification the even algebra is $M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$. Its two irreducible modules are the complex two-dimensional half-spin modules, carrying the two factors of $\mathrm{Spin}(4,\mathbb{C}) \cong \mathrm{SL}(2,\mathbb{C}) \times \mathrm{SL}(2,\mathbb{C})$ [5]. This does not split the generator space into three spatial directions and one time direction, and no physical observer/state interpretation of the two factors is assumed.

Step 4b (cross-reference: the status of the auxiliary $\mathrm{Spin}(4)$ construction). Proposition S3.1 above records this: C4-as-factorisation, in the standard group-theoretic sense of a direct product decomposition into commuting normal subgroups, is not implied by C4a or C4b and does not imply either of them; it is a separate, post-selection algebraic property of the Clifford/spin structure built once $n = 4$ is already fixed.

Steps 1–5 establish Proposition S3.1. They are not used in the proof of Theorem 3.1, which rests on the counting bound (Theorem 3.0) and the explicit witness P_4; the values $n = 1, 2, 3$ are excluded by Theorem 3.0 alone.

S5 Experimental status and falsifiability

The central result of the present work — $n = 4$ as a confirmed admissible representation dimension, and the unique minimal such value ($n = 1, 2, 3$ all excluded; Theorem 3.1; §S3.4), giving $d_{\text{eff}} = 4$ under Theorem 3.1a — is structural rather than dynamical. It does not directly predict the outcome of a specific experiment. The falsification targets are distinct. A C4b counterexample with generator rank below four would refute the counting theorem; failure of the Atomic-Generation identification in a C4b monoid would refute Theorem 3.1a. A genuinely fundamental elementary carrier satisfying C1, C2, C4a, and C4b but realised above the attained minimum would instead falsify C3. A composite rank above four falsifies none of these claims, because joint composite rank is a different invariant (Appendix C).

If an admissible operational coarse-graining exists for a higher-dimensional physical theory, by Theorem 3.1a under C4b its C3-selected target has d(P) = 4; the existence of such a reduction is not established here (Open Problem B.0, Appendix B). This prediction connects the present structural result to the compactification problem addressed in Appendix B, where the reduction of higher-dimensional structures to $d_{\text{eff}} = 4$ is analysed as an operational coarse-graining rather than as a geometric compactification.

S6 Open questions

The open questions are: reconstruction of the enriched local action from categorical data alone (Open Problem A.2); existence of the operational coarse-grainings in Appendix B; attainment of the non-normalising rank-seven composite bound in Appendix C; and physical realisations of the specified grading, observer action and reduced quantum-operation maps. C5 expresses a non-return condition, but neither calibrated time nor persistence of non-factorisation follows from it. The auxiliary spin factorisation is a standard consequence of its added Clifford structure, not a solution of these problems.

S7 Requirement C5 and its relationship to $\boldsymbol{A}([\boldsymbol{s}])$

The present work establishes a minimum achievable $n = 4$ in the class satisfying C1, C2, C4a, and C4b, with C3 selecting that minimum; the Atomic-Generation Theorem then identifies the selected n with d_{eff} under C4b. A natural question arises: if a fifth requirement C5 is added, does d_{eff} change? We now formulate C5 precisely and show that it does not affect $d_{\text{eff}} = 4$.

Requirement C5 (Directed Operational Non-Undoability). Let $P([s]) \subseteq P$ denote the subset of all admissible procedures applicable at indistinguishability class $[s]$. There exists at least one admissible procedure $P* \in P([s])$ — taking $[s]$ to some class $[s_1]$ — such that no finite admissible continuation path $[s_1] \to Q_1 [s_2] \to Q_2 \to \ldots \to$ [sk] in C_D, built from procedures Qi that are each admissible in $P([si])$ applicable at each successively reached class, returns to the initial class $[s]$. Such a procedure is called irreversible. C5 is an additional dynamical condition on accessibility. It does not alter the definition $n := d(P([s]))$, and by itself it does not select a metric signature, a negative Clifford square or a preferred basis generator. The informational content of C5 is directed non-undoability, which may be an ingredient of a later physical time construction but is not itself a claim of causal-temporal ordering: the irreversible procedure $P*$ itself cannot be undone by any admissible continuation, in the sense above. This supplies directed non-undoability, one structural ingredient relevant to persistent memory and quantum entanglement of separated systems (Appendix C); it does not by itself guarantee that the correlations $P*$ creates cannot be erased by some other, later admissible procedure — that further persistence question is addressed separately in Appendix C (Proposition C.2).

Physical motivation for C5. Consider a model in which every admissible physical process has an admissible inverse returning the system to its prior operational class. In such a model no process is required to produce an operationally non-undoable record. C5 excludes this fully reversible limiting case by requiring at least one admissible procedure for which no finite admissible continuation returns to the initial class. C5 therefore supplies directed operational non-undoability that may be relevant to a later theory of records or physical time. It is not claimed here to be a necessary or sufficient condition for a macroscopic arrow of time; the decomposition 4=3+1 and physical time remain outside the scope of this paper.

Relationship to $A([s])$. Requirement C5 does not, by definition, add a fifth generator or require the irreversible procedure P* itself to belong to a chosen minimum generating set G; the P_4 witness merely supplies one example in which a non-undoable procedure can coincide with a generator. C5 is instead a restriction on the directed accessibility structure: among the already admissible continuations there must exist at least one procedure with no finite return path to the initial class. Consequently, C5 can constrain $O([s])$ and $T_{[s]}$ as accessibility data, but no change of the C3-selected local generator count follows from C5 alone.

Effect of C5 on $n = 4$ and on $d_{\text{eff}} = 4$ under Theorem 3.1a. Theorem 3.1 establishes $n = 4$ as the C3-selected minimum within the class satisfying C1, C2, C4a, and C4b. Adding C5

imposes an additional constraint on accessibility — the existence of at least one non-undoable procedure — but does not by itself require additional generators, nor does it by itself reduce the symmetry group $\mathrm{Aut}(C_{\mathrm{D}})$ to any specific subgroup of $\mathrm{SO}(4)$; see the discussion immediately below.

Connection to operational irreversibility. When C5 is realised, the distinguishability structure acquires a preferred directed process — the irreversible procedure $P*$ of C5's definition. C5 introduces this preferred directed operational process, but no symmetry reduction of the form $\mathrm{SO}(4) \to \mathrm{SO}(3,1)$, or even a specific stabiliser subgroup of $\mathrm{SO}(4)$, follows from C5 alone. Such a statement would require an additional rule relating operational irreversibility to a bilinear form; scaling is not an orthogonal transformation, so no group of the form $\mathrm{SO}(3) \times \mathbb{R}^{+}$ is generated by C5 within the orthogonal representation constructed in §S4 either.

In the initial left regular P_4 model, the positive generator increases the progress coordinate and every admissible continuation leaves it unchanged or increases it further. It therefore satisfies C5. The four-generator lower bound remains attainable after this additional accessibility condition is imposed.

Summary. C5 identifies at least one directed, non-undoable process among the admissible procedures. It does not create new algebraic structure and does not by itself fix a metric, signature, or symmetry group beyond the structure already specified before C5. The C3-selected minimum $n = 4$ within the C1, C2, C4a, C4b class remains available after imposing C5 whenever the chosen witness/realisation also satisfies C5; C5 itself supplies no algebraic reason to change the generator lower bound or the Atomic-Generation identification.

Structural necessity of the ordering Proposition S3.1 before C5, and its resolution. A potential objection arises: if C5 is applied and further real-form structure is considered, does Proposition S3.1's auxiliary construction cease to hold? The answer is no, for the following reason. Proposition S3.1's auxiliary construction is formulated at the level of the complexified group $\mathrm{Spin}(4,\mathbb{C}) \cong \mathrm{SL}(2,\mathbb{C}) \times \mathrm{SL}(2,\mathbb{C})$. This decomposition is a property of the complex Lie algebra $\mathfrak{so}(4,\mathbb{C}) \cong \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$, independent of the choice of real form. This algebraic fact, however, does not imply that any particular real form itself factorises into two independent $\mathrm{SL}(2,\mathbb{C})$ groups, and no such claim is needed here: C5 does not disturb Proposition S3.1, since it is a statement about the complex envelope $\mathrm{Spin}(4,\mathbb{C})$, which is unchanged by any further structure C5 might motivate; C4a and C4b are, in any case, independent of this construction, as stated in §3.2.6.

Relationship between C5 and entanglement. The static framework C1–C4 can represent a joint composite state with persistent operational correlations whether the subsystems are currently interacting or already separated; it does not encode a physical time history explaining how those correlations were produced or how long they persist. Structural non-factorisation of $A([s]\mathrm{AB})$ is therefore a property of the present joint accessibility structure, not a statement that entanglement exists only during active interaction. C5 adds the existence of at least one directed, non-undoable procedure, which may be relevant to records or persistence but is not necessary or sufficient for quantum entanglement in general. Appendix C treats persistence with a separate local non-factorisation persistence condition. Persistent quantum entanglement, Bell violation and microscopic irreversibility are not derived from C5 here.

S8 Algebraic consequences and falsifiability

The following consequences have separate hypotheses. Four-generator closure follows from the selected minimum. Clifford and spin statements require the added representation structure. Effective rank reduction concerns a post-procedure monoid that may leave C4b.

S8.1 Consequence 1: four canonical generators under C4b and C3

In the C3-selected C4b class, $E(P)$ is the unique minimum generating set and has four elements. Every procedure is a word in those elements. Moreover, every generating set contains $E(P)$; any additional member is redundant. Thus no larger irredundant generating set exists in this class. This stronger conclusion uses atomic generation, not merely the definition of minimum generator number.

S8.2 Consequence 2: Clifford algebra dimension (conditional on adding a representation structure)

Once a linear representation space $V_{\mathrm{G}} = \mathbb{R}\mathrm{G}$ and a quadratic form q are introduced on it (Appendix A.2.2) — additional structure beyond the bare monoid $P([s])$ — the associated real Clifford algebra $\mathrm{Cl}(V_{\mathrm{G}}, q)$ has dimension $\dim_{\mathbb{R}} \mathrm{Cl}(V_{\mathrm{G}}, q) = 2^4 = 16$ [5]. As a vector space, $\mathrm{Cl}(V_{\mathrm{G}}, q)$ has a one-dimensional scalar subspace $\mathbb{R} \cdot 1$ (not a two-sided ideal, so no algebra quotient is intended); the quotient vector space $\mathrm{Cl}(V_{\mathrm{G}}, q)/\mathbb{R} \cdot 1$ has dimension 15. This numerically coincides with the $4^2 - 1 = 15$ real parameters of a two-qubit density matrix in standard quantum mechanics. We record this as an algebraic coincidence, not a derivation of two-qubit state tomography: the correspondence between $\mathrm{Cl}(V_{\mathrm{G}}, q)$ and a Hilbert-space density-

matrix structure requires an additional structural bridge — a way of identifying algebra elements with physical tomographic operators — that is not constructed in this paper.

S8.3 Consequence 3: spinorial weight in the $\mathbf{Spin(4)}$ realisation (conditional on the complementary-support construction of §S4)

In the $\mathrm{Spin}(4)$ complementary-support consistency construction of §S4 — an oriented Euclidean spin structure added on top of $n = 4$ — $\mathrm{Spin}(4) \cong \mathrm{SU}(2)_{\mathrm{L}} \times \mathrm{SU}(2)_{\mathrm{R}}$ (Theorem A.4). The fundamental representations of each $\mathrm{SU}(2)$ factor are spinorial, of weight 1/2. This makes a spin-1/2 structure canonically available within this added representation once it is constructed; it does not by itself derive the existence of physical fermions, nor that nature must realise this structure, nor is 1/2 obtained as a 'topological index' or dimension ratio of the two factors — $\dim \mathrm{SL}(2,\mathbb{C})_{\mathrm{L}}$ and $\dim \mathrm{SL}(2,\mathbb{C})_{\mathrm{R}}$ are both 3, and their ratio is a separate, unrelated quantity from the representation weight 1/2.

S8.4 Consequence 4: conditional rank-reduction bound under measurement

For a structure-reducing procedure P, define $\Delta n = n([s]) - n_{\mathrm{eff}}([s']; P) > 0$, using the effective target monoid of Appendix C. If the source has rank four, the effective target cannot satisfy C4b. If that effective target is nontrivial, its rank is at least one and $1 \leq \Delta n \leq 3$; without nontriviality a rank-zero target is possible. The fundamental monoid at the target is a different object. This statement concerns generator rank and does not establish a quantum outcome law.

S8.5 Remark on compactification

If a compactification scheme (Appendix B) is specified independently, its holonomy and bundle data are additional input to the present framework; no gauge structure is derived here.

S8.6 Summary

Four-generator closure follows from the selected minimum; uniqueness of the minimum generating set uses atomic generation. The rank-reduction statement assumes a specified effective target monoid and an actual decrease of generator number. Clifford dimension and the spin representations require the extra representation structures of §§S8.2–S8.3. None of these statements identifies an operational generator rank with a quantum state-space dimension.

S8.7 Relation to familiar quantum structures

Consequences 1, 2, and 4 of §S8 coincide numerically with results familiar from standard quantum mechanics: four-generator closure, 15 parameters for two-qubit tomography once a representation structure is added, and integer rank of the post-measurement state. The derivations, however, proceed from different starting points.

In quantum mechanics, these numbers are consequences of structural postulates that are introduced without derivation: the number 4 follows from postulating a two-dimensional complex Hilbert space $\mathbb{C}^2$ ($\dim\mathbb{R} = 4$); the number 15 follows from postulating the tensor product $\mathbb{C}^2 \otimes \mathbb{C}^2$ and the trace normalisation; the integer rank follows from postulating the matrix algebra structure of density operators. Some structures obtained after adding the representation-theoretic extensions of §§S8.2–S8.3 exhibit numerical coincidences with these familiar quantum constructions. These coincidences should not be interpreted as identifications of $d(P)$, Hilbert-space dimension, density-matrix dimension, or spacetime dimension: the numbers are equal, but no theorem relating these distinct notions of dimension is proved here.

In this framework, C4b gives the lower bound and an explicit witness attains four; C3 separately selects that minimum. No Hilbert space, tensor product or complex field is needed for this counting argument. Sections S8.2–S8.3 add representation structures explicitly and do not reconstruct quantum mechanics from the procedure monoid.

This correspondence appears once the additional structure used by quantum mechanics is introduced on top of the present framework; it does not show that the quantum postulates are forced by operational distinguishability.

For comparison, orthomodular and orthogonality-space approaches form a distinct line of reconstruction [8–12].

A derivation of a quantum outcome-probability rule would require additional assumptions and is not attempted here. The present algebraic results and their record-set formulation do not assume a probability measure.

Corollary S8.1 (Operational prohibition bounds). The following constraints on physically admissible processes hold within the scopes stated individually below, collected here as a single statement for clarity:

(i) Four-generator closure. No fifth generator is required to generate $P([s])$ in the C3-selected minimal class: any admissible physical procedure reduces to a finite composition of the four minimal generators $\{g_1, g_2, g_3, g_4\}$. By the Atomic-Generation Theorem under C4b, this implies that no fifth operationally independent distinguishability direction exists.

(ii) Rank reduction leaves the C4b class. A structure-reducing procedure applied to a carrier with $n = 4$ produces an effective monoid outside C4b (Theorem 3.0); if that monoid satisfies C1, the generator rank drops by between one and three (§S8.4). The operational-rank identification is not claimed for the target.

(iii) No fractional generator rank. Under Axiom G, $n = d(P([s]))$ is the cardinality of a finite minimum generating set and is therefore an integer. Under C4b, Atomic Generation transfers this integrality to $\mathrm{rank}\, A([s])$. Thus changes of these algebraic ranks, whenever both the initial and final quantities are defined within the corresponding scope, occur in integer steps; a value such as 3.7 is not a value of these invariants.

(iv) Auxiliary Spin-algebra invariant. After the optional $\mathrm{Spin}(4,\mathbb{C})$ carrier is introduced, its Lie algebra has the semisimple decomposition $\mathfrak{so}(4,\mathbb{C}) \cong \mathfrak{sl}(2,\mathbb{C})L \oplus \mathfrak{sl}(2,\mathbb{C})R$, unique as an unordered pair of simple ideals. An automorphism may exchange the two isomorphic ideals but cannot replace this simple-ideal decomposition by a genuinely mixed one. This is an algebraic invariant of the added Spin carrier. It is not a prohibition derived from C4 on arbitrary physical procedures, and it does not assert that operational observer transformations act as arbitrary automorphisms of $\mathfrak{so}(4,\mathbb{C})$.

The scopes of (i)–(iv) differ and should not be conflated. Statement (i) follows from C3 together with Theorem 3.1. Statement (ii) concerns the generator rank of a post-procedure effective monoid that, by Theorem 3.0, lies outside C4b. Statement (iii) follows from Axiom G for n and from C4b for operational rank. Statement (iv), by contrast, is conditional on adding the auxiliary $\mathrm{Spin}(4,\mathbb{C})$ structure and is only an invariant of that added semisimple Lie algebra; failure to realise the optional Spin carrier would not falsify Theorem 3.1a. The direct operational falsification criteria are therefore those tied to the monoid/rank statements within their stated assumptions, not the existence of the optional Spin construction.

S9 The logical chain from distinguishability to information

The construction of the present work makes explicit a logical chain from distinguishability to information, developed in detail in §S2: physically distinguishable states exist as the primitive setting → the stated structural class constrains the minimum independent procedure rank → C1 supplies non-trivial local accessibility → typed morphisms represent admissible transformations → outcome-registering morphisms represent measurements → distinguishable records carry operational information. The formal content of this chain is given in §S2.2–S2.8. This chain provides a structural foundation for physics, not a derivation of quantum mechanics in full: the framework does not contain probability amplitudes, interference, or quantum phases. These require additional structure beyond the present operational framework.

This chain keeps the logical statuses distinct: distinguishability and multiplicity belong to the primitive operational setting, while accessibility and measurement are introduced through the stated structural requirements and operational definitions; information is then derived from the existence of distinguishable outcome records.

S10 Relation to higher-dimensional theories

Corollary 3.2 compares procedure monoids with different generator numbers. If a higher-rank monoid admits a surjective reduction to a C3-selected admissible target, that target has rank four. Existence of such a reduction is an additional question (Appendix B). No conclusion about a higher-dimensional spacetime theory follows without a separate identification of its procedures.

Higher-dimensional theories such as string theory and M-theory work with ten or eleven spacetime dimensions and recover four effective dimensions through compactification [6, 7]. The present results do not identify spacetime dimension counts with the operational generator rank $d(P)$. Corollary 3.2 concerns operational structures with more than four generators only; whether compactification schemes correspond to admissible operational reductions in the sense of Appendix B is an open question (Open Problem B.0).

S11 On the word 'cause'

The paper asks for an attained minimum generator number under specified assumptions. This is a structural question, not a dynamical explanation of why physical states exist.

The assumptions define an admissible class; the counting theorem determines its generator lower bound. Their physical applicability requires a realisation and is not implied by the algebraic proof.

S12 A proposed structural encoding of phase transitions

S12.1 Phase transitions in the language of distinguishability

For an auxiliary graph description, fix an ambient set of operational classes and a declared control protocol. Draw a directed edge when a procedure allowed by that protocol connects two classes. Any claim about components must specify directed or undirected connectivity and the vertex set. The full forward-reachable graph including its root is weakly connected by construction; the identity also makes the root reachable from itself. A nontrivial diagnostic therefore requires a separately specified restricted graph, for example after excluding a reference class. This restriction is not determined by C4b.

S12.2 First-order transitions: a proposed structural analogue of loss of connectivity

A restricted graph can have more than one component, but graph disconnection is not by itself a criterion for a first-order transition. A thermodynamic correspondence would require a concrete model relating the restriction to phase coexistence, control constraints and free energy. No such correspondence is proved here.

S12.3 Second-order transitions: a proposed structural analogue of change of symmetry

A second-order phase transition may, in a future model, be represented structurally by a change in a specified symmetry group acting on the relevant local accessibility structure while connectivity is retained. The present framework does not prove that the full categorical $\mathrm{Aut}(C_{\mathrm{D}})$, or even the local observer stabiliser, must undergo the Landau symmetry-breaking pattern at a critical point. The intended statement is therefore a candidate correspondence: if a concrete operational model exhibits such a symmetry change, an order parameter may be associated with an invariant that distinguishes the two symmetry regimes. No Landau order parameter is derived from $A([s])$ here.

S12.4 Ideal gas versus real gas

The ideal-gas comparison is likewise only a proposed diagnostic, not a theorem. A concrete operational model of an ideal gas would have to show that its relevant accessibility structure

remains in the same connectivity and symmetry class throughout the thermodynamic parameter range; the absence of intermolecular interactions does not, by itself, prove this statement in the abstract monoid formalism. The framework can represent such a no-transition result once the model-specific procedure structure is supplied, but it does not derive the thermodynamics of the ideal gas from C1–C4 alone.

For a real gas, phase coexistence and metastability require model-specific thermodynamic input. Interactions alone do not imply disconnection of an operational graph, and no coexistence curve is calculated from the procedure monoid.

S12.5 Nucleation

A concrete model can distinguish configurations with and without a nucleus through specified tests. The word local in the monoid construction refers to a forward domain, not automatically to a spatial region. Identifying a nucleus requires additional physical localisation data.

A model may distinguish configurations containing a nucleus from those without one. Critical size, growth and decay require additional dynamics. C5 alone neither supplies those dynamics nor is proved necessary for them.

S12.6 Connection to the companion framework and C5

Phase-transition dynamics, including nucleation, require additional model-specific dynamical and statistical structure beyond the static results proved in this paper.

S12.7 Scope of the proposed encoding

The following paragraphs distinguish the proposed structural encoding from thermodynamic claims.

A phase transition is a change in the physical organisation of a system — for example, water freezing into ice, or a ferromagnet crossing its Curie temperature. Standard thermodynamics describes such transitions through free energies, equilibrium ensembles, symmetry and fluctuations. The present framework does not replace that machinery. Section S12 asks only whether changes in an independently specified operational accessibility structure could provide a useful structural representation of some features of phase transitions once a concrete physical model supplies the missing thermodynamic and dynamical content.

In this proposed representation, a first-order transition could correspond to a change in connectivity or accessibility between operational sectors associated with coexisting phases. Such a correspondence must be demonstrated in a concrete model; it is not a theorem of C1–C4. A free-energy barrier, latent heat, nucleation rate, or energetic cost is additional thermodynamic information and is not obtained from graph disconnection alone.

For a continuous transition, a concrete model might instead encode the transition through a change of a specified symmetry acting on the accessibility structure while connectivity remains intact. This is only a structural analogue of spontaneous symmetry breaking: the present theory does not derive the Landau order parameter, critical exponents, or the Curie temperature from $\mathrm{Aut}(C_{\mathrm{D}})$, a local stabiliser, or $\mathrm{rank}\, A([s])$.

The ideal-gas comparison should also be read conditionally. If a concrete operational model of an ideal gas retained the same relevant connectivity and symmetry class throughout its thermodynamic parameter range, the framework would encode the absence of a transition by that structural stability. The absence of intermolecular interactions alone does not prove this property inside the abstract monoid formalism, and a real gas is not shown here to acquire a transition merely because interactions are present.

Nucleation can likewise be represented only after a model specifies what counts operationally as a local region in the new phase. One may then describe a nucleus as a local domain whose accessibility or symmetry pattern differs from that of its surroundings. This is a candidate structural encoding of a nucleus, not a derivation of critical-nucleus size, surface tension, activation free energy, or nucleation kinetics.

The present framework does not yet describe the dynamics of these processes — how fast transitions occur, how nuclei grow, or what drives the system from one phase to another. Those questions require additional dynamical and, for rates and thermal activation, typically probabilistic or thermodynamic structure. C5 may be useful as one condition expressing directed non-undoability, but the present work does not establish C5 as necessary or sufficient for phase-transition dynamics, nor does it derive a physical time variable from it.

S12.8 Directions for future work: magnetic phase transitions, nucleation rates, and broader applicability

Unified description of magnetic phase transitions in zero and finite field. A natural and concrete application of the present framework, left for future work, concerns magnetic phase transitions in

an external field — specifically, the question of whether the sharp Curie point at $H = 0$, the smeared thermodynamic anomalies observed at finite H, and the eventual disappearance of any distinguishable transition at saturation fields can be described as a single structural phenomenon rather than three separately fitted regimes, as in the Landau–Belov–Arrott approach.

A field-dependent family of models may be denoted $\mathcal{C}_D(H)$, with H an externally specified real control parameter. This notation does not introduce a topology on the observer group. Any continuity of the family in H, and any relation between its categorical invariants and thermodynamic response, must be specified independently.

For a magnetic model, one must distinguish symmetry of the laws and allowed procedures from symmetry of a selected phase or state. Spin-reversal symmetry at zero field is not generally absent above the Curie temperature; a selected ordered phase can break a symmetry of the underlying model. Neither finite field nor saturation alone implies a loss of exactly one procedure generator. Such a rank change would have to be demonstrated from the specified effective monoid.

A future quantitative model built on the family $\mathrm{C_D}(H)$ could differ from Landau theory [13] in how it organises its primitive variables, but the present paper does not compute the magnetisation M, the transition temperature Tc, or their field dependence from $\mathrm{Aut}(\mathrm{C_D}(H))$ or $\mathrm{rank}\, A([s])$. Those quantities require model-specific energetic, statistical and dynamical input. Two further points remain explicit: the saturation field at which any proposed rank change occurs is not fixed by distinguishability alone, and smooth dependence of the family $\mathrm{C_D}(H)$ on H is an additional physical assumption unless derived from a microscopic model. Thus §S12 records a structural research programme, not a completed alternative thermodynamic theory.

Nucleation rates and critical sizes require a physical dynamical model and thermodynamic data. C5 supplies no calibrated time or rate. The graph language above is therefore only an optional encoding, not an alternative calculation of nucleation.

Extensions to topological phases or nonequilibrium systems require separate operational models. Accessibility-graph connectivity is not a substitute for a band invariant, and generator rank is not a thermodynamic order parameter. These possible applications are not used as evidence for the present theorems.

S13 Mathematical and physical scope

This section collects the scope distinctions used in the main text and the supplementary constructions.

Scope distinction 1. The Clifford/spin construction is optional. It follows only after a quadratic form is added to a frame space and does not classify all operational observer groups.

Scope distinction 2. The faithful representation is of the setwise stabiliser of a generating set in a local monoid. A general categorical automorphism need not act on that monoid; the admissible observer maps explicitly preserve the enriched action.

Scope distinction 3. The rank–generator equality follows from atomic generation under the stated grading and finite zero-sector hypotheses. It does not equate operational rank with spacetime dimension.

Scope distinction 4. C4a records isomorphism invariance. The numerical lower bound uses the separate assumptions of C4b: a nontrivial grading, frame regularity and a faithful non-Abelian action on the canonical zero-grade generators.

Scope distinction 5. The procedure-monoid invariant differs from state-space dimension. General process theories need not assume Hilbert spaces; no claim that this framework precedes or subsumes all categorical approaches is made.

Scope distinction 6. A mathematical counterexample satisfying the theorem hypotheses would refute a theorem. An independently identified elementary physical realisation above the attainable minimum would instead challenge C3. The paper supplies no new numerical prediction distinguishing quantum mechanics from experiment.

Scope distinction 7. Atomic generation is proved using positive integer degree and finiteness of the zero sector. The positive-height-free counterexample in the main text shows why the stated grading hypothesis matters.

Scope distinction 8. Proposition S3.1 records the standard complex spin factorisation after rank four and a quadratic form have been supplied. It is not part of the counting argument.

Scope distinction 9. The complex spin group is an added representation object. Its factorisation does not select a real form, Lorentzian signature or physical time.

Scope distinction 10. The enriched action is a domain with an acting procedure monoid. It is not an observable algebra or the full category of quantum channels. The quantum maps in main-text Example 3.3 are an independently specified target of a homomorphism.

Scope distinction 11. Axiom F bounds the distinguishable outputs of each procedure on each covering class. It does not impose a finite global alphabet. Continuous-outcome instruments require an explicitly chosen finite-resolution restriction if they are to be included.

Scope distinction 12. A common family of deterministic readouts defines a transitive equality-of-responses relation. Nontransitivity here is permitted by the pair-dependent comparison domains. Agreement under two different individual tests alone does not establish the nontransitivity of indistinguishability. The category uses typed maps on source representatives; the full local monoids are retained separately.

Scope distinction 13. Structural non-factorisation also occurs classically. The copying map and diagonal domain in main-text §3.4.3 are explicit examples. Consequently this criterion is neither a quantum-entanglement witness nor a criterion for Bell nonlocality.

S14 Remarks on the selected value $n = 4$

The value four selected here is an operational rank. Riemannian geometry is formulated for arbitrary dimension [15]; general relativity is formulated in four spacetime dimensions [16]; higher-dimensional theories obtain four effective dimensions through compactification. The present result concerns a different quantity — the minimum generator rank within the stated class of graded operational procedure monoids — and no identification with spacetime dimension is made. If a physical system failed to satisfy the admissibility conditions C1, C2, C4a, C4b or the selection principle C3, this would place it outside the class considered here; it would not affect the counting theorem. After $n = 4$ is selected, the auxiliary Clifford/spin carrier gives the familiar decomposition $\mathrm{Spin}(4,\mathbb{C}) \cong \mathrm{SL}(2,\mathbb{C})_{\mathrm{L}} \times \mathrm{SL}(2,\mathbb{C})_{\mathrm{R}}$ (§S4); this is recorded as an algebraic consequence and is not used in the proof.

Appendix A. Formal framework

A.0a Observer sector

The observer sector is the subgroup generated by specified admissible state bijections whose inverses are admissible and whose conjugations preserve local procedure monoids. This is the definition in main-text §3.2.1a; no connected topology or determinant restriction follows.

A.1 Finite generator number

Theorem A.1. Under Axiom G, $d(P) \leq |G| < \infty$ for a finite generating set G. Proof. The minimum cannot exceed the cardinality of a given generating set. ▫

A.2.1 Typed category and enriched actions

Morphisms are the typed representative maps defined in main-text §2.4.9. Equality of maps with fixed source and target is a congruence for composition, proving the category axioms. The full procedure action on a forward domain is additional data and need not be recoverable from this category.

A.2.2 Frame representation

For a minimum generating set G, define $V_G = \mathbb{R}^G$. Theorem A.2. The setwise stabiliser $H_G \leq \mathrm{Aut}(P)$ acts faithfully by permutation matrices on V_G. Proof. An element of the kernel fixes every generator and hence every word, so is the identity. ▫

A.3 Invariant inner product

Theorem A.3. The representation in Theorem A.2 preserves a positive-definite inner product. Proof. Declare the frame basis orthonormal. Every permutation matrix is orthogonal. Equivalently average any positive-definite form over the finite group H_G. Orientation preservation requires a separate restriction. ▫

A.4 Optional spin construction

Theorem A.4. After choosing an oriented positive-definite quadratic structure on a four-dimensional frame space, its complex spin group is $\mathrm{Spin}(4,\mathbb{C}) \cong \mathrm{SL}(2,\mathbb{C}) \times \mathrm{SL}(2,\mathbb{C})$. The even complex Clifford algebra is $M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$; its two half-spin modules give the two factors

(§S4, [5]). This is an auxiliary representation-theoretic fact, not a decomposition of the generator space into space and time.

Primary logical structure (main text §3.2.6a–b; establishes Theorem 3.1 within the stated C1, C2, C4a, C4b class, independent of this Appendix): Requirement C4b (§3.2.6: a procedure grading, $R_1 + R_2$ frame regularity, and non-Abelian observer richness) → Theorem 3.0: $n \geq 4$, by a representation-free counting argument. The explicit finitely generated monoid $P_4 = \mathbb{N}_0 \times B_3 \rightarrow d(P_4) = 4$. Theorem 3.0 + P_4 + requirement C3 → Corollary 3.0 / Theorem 3.1: $n_{\min} = 4$.

Supplementary logical structure (this Appendix — a post-selection algebraic construction on a different carrier from the value already established above, not needed to establish it, and not a check on it):

Axiom G (§2.1.2) → Theorem A.1 (finiteness of the representation rank): $d(P([s])) < \infty$. Axiom F is not needed for this algebraic implication. Hence n is well-defined; under C4b, the Atomic-Generation Theorem also gives $\operatorname{rank} A([s]) = n$.

Axiom G and a chosen minimum generating set G imply the local stabiliser representation of Theorems A.2–A.3: the setwise stabiliser $\mathrm{Aut}(P([s]))G$ acts by permutations of G, its linear extension is faithful on the free frame space $\mathbb{V}([s]) = \mathbb{R}\mathrm{G}$, and finite averaging places its image inside $O(n)$. This is a representation of the local stabiliser, not of the full $\mathrm{Aut}(\mathrm{C_D})$. An $\mathrm{SO}(n)$ statement is available only after an explicit orientation-preserving restriction; identity reachability alone does not force determinant $+1$. These facts are auxiliary to, and not used in, the representation-free counting theorem.

Appendix A.4 records a separate post-selection construction after $n = 4$ is already fixed. Once a positive-definite quadratic form and the associated Clifford/spin carrier are introduced, the complexified group admits $\mathrm{Spin}(4,\mathbb{C}) \cong \mathrm{SL}(2,\mathbb{C})_{\mathrm{L}} \times \mathrm{SL}(2,\mathbb{C})_{\mathrm{R}}$ on the corresponding half-spin modules. This is not a decomposition of the generator space $\mathbb{V}([s])$, not a theorem about the full categorical $\mathrm{Aut}(C_{\mathrm{D}})$, and not a second proof or check of C4a, C4b, or $n_{\min} = 4$. Its role is solely to record a familiar chiral algebraic structure available at the selected value four.

The minimum $n_{\min} = 4$ is established in the main text by Theorem 3.0 plus the explicit four-generator witness and C3; the auxiliary Spin case analysis is retained only as a post-selection representation-theoretic discussion and is not used to exclude candidate ranks.

The atomic-generation theorem then gives rank $A([s]) = d(P([s]))$ under C4b, so for the C3-selected fundamental local carrier $d_{\text{eff,fund}} = 4$.

Open problem A.2: is there a canonical categorical reconstruction of the enriched local action $A([s]) = (D([s]), P([s]) \curvearrowright D([s]))$ from the typed morphism structure of $\mathrm{C_D}$ alone, without taking $P([s])$ as separately supplied local data? No colimit or Kan-extension construction is claimed; this is stated only as an open question. It does not affect the main result. The separate auxiliary representation $\mathbb{V}([s]) = \mathbb{R}G$, the free real vector space on a chosen minimum generating set G, is established independently (Appendix A.2.2) and is not itself part of this open question.

Appendix B. Compactification as operational coarse-graining

Appendix B specifies an additional reduction problem; existence is not asserted by the counting theorem.

An operational coarse-graining from an N-generator operational structure to a four-generator target is a typed functor $CG : CD_N \to CD_4$ preserving sources and targets, identity morphisms and composition, such that the target satisfies C1, C2, C4a and C4b and its target local monoids have generator rank $d(P) = 4$. In addition, for every object [s] the functor must induce a surjective monoid homomorphism from the local monoid at [s] onto the local monoid at CG([s]), compatible with the actions on the forward domains. Without this condition the question would be trivial, since a constant functor onto a single object with identity arrows satisfies the remaining clauses. Surjectivity on objects is not required. Preservation of the optional auxiliary $Spin(4,\mathbb{C}) \cong SL(2,\mathbb{C})L \times SL(2,\mathbb{C})R$ carrier, where that extra structure has been introduced, is a separate optional condition and is not part of C4. Rank reduction need not mean that exactly $N - 4$ chosen generators are sent to the identity: it may arise through generator identifications, new relations, or images that are themselves composites. Open Problem B.0 is existence: Theorem 3.1 does not construct such a reduction from an arbitrary $N > 4$ structure. Comparisons with specific higher-dimensional theories would require an additional physical bridge and are not made here. Compatibility and non-uniqueness remain additional questions conditional on existence.

Appendix C. Measurement, entanglement, and composite systems

Effective and fundamental monoids. For an admissible procedure P: $[s] \to [s']$, let $P_{\text{eff}}([s']; P)$ be a specified monoid of accessible post-procedure continuations and set $n_{\text{eff}} = d\big(P_{\text{eff}}([s']; P)\big)$. It is distinct from a fundamental target monoid to which C3 may apply. Call the procedure structure-reducing when $n_{\text{eff}} < d\big(P([s])\big)$. A decrease below four leaves C4b; equality of operational and generator ranks at the target requires the Atomic-Generation hypotheses there.

Proposition C.1a (a sufficient rank-reduction mechanism). Let G be a minimum generating set of $P([s])$ and let P_*: $P([s]) \to P_{\text{eff}}([s']; P)$ be a homomorphism. If a proper subset $G' \subset G$ has images generating the effective target, then $d(P_{\text{eff}}) \leq |G'| < |G|$. In particular this holds if P_* is surjective and every generator outside G' maps to the identity. Proof. The images of G' provide a generating set of the stated size. ▫ Existence of such an induced map is an additional assumption; it does not follow from measurement alone.

The composite-system results — the product rank $d(P_{\text{A}} \times P_{\text{B}}) = d(P_{\text{A}}) + d(P_{\text{B}})$ (Lemmas 3.1–3.2), the lower bounds under full local completeness (six in general, seven for two copies of P_4 or when the product of the two local observer symmetries is preserved; Theorems 3.2–3.4), the classification of the zero-grade sector at $n = 4$ into the types M3 (Type I) and B3 (Type II) (Lemmas 3.3–3.4), the bounds $3m + 1 \leq n_{\min}(m) \leq 4m$ for m subsystems (Theorem 3.5), the non-factorisation criterion (Proposition 3.1), the detection of joint procedures by non-local canonical generators (Theorem 3.6) and a sufficient preservation criterion for domain non-factorisation (Proposition 3.2), a conditional bridge to tensor non-separability (Proposition 3.3), the commutativity of idempotent procedures and its consequences for non-commuting quantum operations (Proposition 3.4, Corollary 3.5, Theorem 3.7, Proposition 3.5), and the record-locality criterion excluding Hardy-type obstructions (Theorem 3.8, Lemma 3.5) — are stated and proved in §3.4 of the main paper. The material below complements them.

Rigidity of the hypothetical $n = 7$ case. If, in addition, $d(P_{\text{AB}}) = 7$ for two copies of the paper's own witness P_4 (so R_A, R_B are both Type II), the argument of Theorem 3.3 of the main text forces $R_A \times R_B \subseteq R_{\text{AB}}$ to be onto (since $|R_A||R_B| = 64 = 2^6$ already saturates the $|R_{\text{AB}}| \leq 2^{\text{k}}$ bound at $k = 6$), so $R_{\text{AB}} \cong R_A \times R_B \cong B_6$ exactly, and $|J(R_{\text{AB}})| = 6$ exactly. By the Atomic-Generation Theorem, P_{AB} then has exactly one positive composition-irreducible generator τ: $E(P_{\text{AB}}) = J(B_6) \cup \{\tau\}$. It must NOT be assumed that the embedded local progress generators τ_A and τ_B (the images of each subsystem's own progress generator under the

embedding) equal τ itself, nor that they carry joint grade 1: since the embedding need not preserve C4b's grading, both τ_A and τ_B may be composite words in τ and zero-grade elements of B_6. Whether such words can be chosen consistently, so that the full embedded copy of $(\mathbb{N}_0 \times B_3) \times (\mathbb{N}_0 \times B_3)$ remains injective, is the open algebraic question addressed below.

Remark (non-product is not the same as entangling). A bipartite SWAP is not a useful counterexample here: it squares to the identity, and Lemma 3.1 of the main text excludes non-trivial invertible elements under C4b. A deterministic example suffices: the classical copying map $h(x, y) = (x, x)$ is idempotent, so it can lie in a zero-grade sector, and it is not a product of local maps (§3.4.3 of the main text). Genuinely non-product procedures and entanglement-creating ones are therefore distinct notions, and nothing established here identifies them.

C3 applies to designated fundamental elementary monoids satisfying its assumptions. The composite results retain an embedded product of the local monoids, and constrain the joint monoid separately. They do not impose rank four on the composite. Changes in physically available local continuations must be modelled explicitly; Theorem C.3 concerns fixed retained local monoids.

Theorem (no rank-seven compression in the normalizing class). Assume the hypothetical rank-seven structure above ($R_{\mathrm{AB}} \cong B_6$, single positive composition-irreducible τ), and additionally assume that τ normalizes R_{AB}: $\tau r = \varphi(r)\tau$ for every $r \in R_{\mathrm{AB}}$ and some semilattice endomorphism φ of R_{AB}. Then P_{AB} cannot contain an injective copy of the two independent progress counters $\mathbb{N}_0^2$ required by $P_A \times P_B$. Proof. Every word containing k copies of τ reduces, under the normalizing relation, to $r\tau^{\mathrm{k}}$ for some $r \in R_{\mathrm{AB}}$. Let the embedded independent positive elements u, v contain p and q occurrences of τ respectively. Set $U := u^{\mathrm{q}}, V := v^{\mathrm{p}}$. Then U and V have equal τ-count pq, and would still freely generate an $\mathbb{N}_0^2$ submonoid if u, v did, since $(m, n) \mapsto U^m V^n$ is injective whenever $(m, n) \mapsto u^m v^n$ is. For fixed N, all $N + 1$ words $U^m V^{N-m}$, for m from 0 to N, have equal τ-count N_{pq} and hence all reduce to $r_{\mathrm{m}}\tau^{\mathrm{Npq}}$ for some $r_{\mathrm{m}} \in R_{\mathrm{AB}}$. For the canonical witness with $R_{\mathrm{AB}} \cong B_6$, only 64 possible tags r_{m} exist. Taking $N = 64$ gives 65 exponent pairs but only 64 possible tags, so two different pairs m give the same element r_{m}, contradicting injectivity of the embedded $\mathbb{N}_0^2$. This argument excludes direct-product, semidirect-product, automorphism, endomorphism, involutive-action, and arbitrary long-code constructions whenever τ can be pushed through R_{AB} by a single endomorphism.

Scope remark: normalization is not derived from C1–C4b. The normalization condition $\tau R_{\mathrm{AB}} \subseteq R_{\mathrm{AB}}\tau$ used in the theorem just proved is not derived anywhere in this paper from C1–C4b; it is an additional structural hypothesis introduced specifically to make the theorem's proof go through. The theorem above is therefore conditional, and must not be read as an unconditional $n_{\mathrm{AB}} \geq 8$ result. A hypothetical rank-seven composite can exist only through genuinely non-normalizing behaviour, in which words $r_0\tau r_1\tau \ldots \tau r_{\mathrm{k}}$ retain positional information that cannot be reduced to a single R-tag times τ^{k}. Current C4b, as stated in this paper, does not exclude this possibility.

Current tight status for the official witness. Full local completeness proves, for $every\ P_{\mathrm{AB}}$ built from two copies of the paper's own witness P_4 under C4b: $d(P_{\mathrm{AB}}) \geq 7$. This is a lower bound on every such P_{AB}, not merely on some constrained minimum — a particular P_{AB} may have rank 8, 9, or higher, if it contains additional independent joint generators beyond what full local completeness alone forces. Define the constrained composite minimum $n_{\min}^{\mathrm{AB}} :=$ $\min\{d(P_{\mathrm{AB}}) : P_{\mathrm{AB}}$ satisfies the B_3/full-local-completeness conditions above$\}$. The lower bound of 7 is Theorem 3.3 of the main text; the upper bound of 8 is attained by the strict product $P_A \times P_B$ itself. Hence $7 \leq n_{\min}^{\mathrm{AB}} \leq 8$. For a hypothetical rank-seven candidate, the additional normalization hypothesis of the preceding theorem is impossible. This excludes rank seven in that normalizing rank-seven class, but by itself does not define or determine a separate normalizing-subclass minimum for arbitrary higher-rank composites. Whether $n_{\min}^{\mathrm{AB}}$ itself equals 7 or 8 — i.e. whether a genuinely non-normalizing composite of rank 7 exists, one in which positional information in interleaved τ-words survives — remains open. This paper does not assert $4 + 4 \rightarrow 7$ as a theorem, and does not assert $4 + 4 \rightarrow 8$ as a theorem for $n_{\min}^{\mathrm{AB}}$: the exact value of this constrained minimum, outside the normalizing subclass, is not established here.

Open problem (non-normalizing rank-seven composite). Does there exist a C4b monoid P_{AB}, generated by the six atoms of B_6 together with one positive composition-irreducible τ, containing $(\mathbb{N}_0 \times B_3) \times (\mathbb{N}_0 \times B_3)$ as an injective sub-monoid, in which τ does not normalize B_6? Equivalently: can positional information in interleaved words $r_0\tau r_1\tau \ldots \tau r_{\mathrm{k}}$ encode two independent, commuting local progress counters? A positive answer would establish $n_{\min}^{\mathrm{AB}} = 7$ for the stated B_3/full-local-completeness class; a general impossibility proof covering the non-normalizing rank-seven case would instead establish $n_{\min}^{\mathrm{AB}} = 8$ for that same constrained class. Neither has been obtained in this paper.

Proposition C.2 (scope of persistence). C5 does not by itself imply preservation of structural non-factorisation under local continuations. To see this at the level of joint domains, take states $(k, x, y) \in \mathbb{N}_0 \times \{0,1\}^2$, admit only non-decreasing changes of k, and include a local reset $x \mapsto 0$. The progress step is non-returning, but the nonrectangular relation $\{(0,0), (1,1)\}$ is mapped by the reset to $\{0\} \times \{0,1\}$. Thus non-returning progress and loss of domain non-factorisation coexist. Conversely, a system restricted to identity procedures preserves a nonrectangular domain while failing C5. These examples concern the logical independence of the accessibility and persistence conditions, not the full fundamental C4b class. A sufficient preservation condition for domain non-factorisation is local injectivity (main-text Proposition 3.2). Persistence of the complete enriched action requires its own continuation condition.

Theorem C.3 (retained local monoids). Suppose two fixed local monoids P_A, P_B have generator rank four and an embedding $P_A \times P_B \hookrightarrow P_{AB}$ retains them as the declared local procedure monoids. Then each embedded local monoid still has generator rank four. Proof. Each factor is isomorphic to its image, and generator rank is invariant under isomorphism. ▫ This statement does not prove that an interaction preserves the physical repertoire of a subsystem. Such a claim requires an identified isomorphism of its pre- and post-interaction local monoids. Effective rank reduction concerns a different continuation monoid, as in Proposition C.1a. The joint rank is governed by the separate composite bounds.

Declaration of generative AI in scientific writing

The author used generative AI tools, including Claude by Anthropic, ChatGPT by OpenAI, and Gemini by Google, to assist with language editing, formatting and notation consistency, reference cross-checking. During the preparation of the substantially expanded later versions of the manuscript, generative AI tools also assisted the author in exploring alternative mathematical formulations and analytical approaches. The research direction, physical problems, and foundational principles of the theoretical framework were defined by the author. The selection of physically meaningful formulations, the interpretation of the results, and the scientific conclusions were determined by the author, who assumes full responsibility for the manuscript.

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